

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2021

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
-

- Let G be a simple group of order 168 (that is, G has no non-trivial normal subgroups).
 - Find the number of elements in G having order 7.
 - Show that G has a subgroup of order 21.
 - Show G has no subgroup of order 14.
- Let $n \geq 1$, let F be a field, and let M be a matrix in $M_n(F)$ ($=n \times n$ matrices with entries in F). Show that there exists a matrix $N \in M_n(F)$ such that $M = MNM$. (*Hint: $F^n = \text{Ker}(M) \oplus V$ for some subspace V .*)
- Let R be a commutative ring with identity, let $n \geq 1$, and let

$$A = R^n = R \oplus R \oplus \cdots \oplus R.$$

We can make A into a ring by defining addition and multiplication componentwise (for example, $(a_i, \dots, a_n)(b_1, \dots, b_n) = (a_1b_1, \dots, a_nb_n)$).

- Let I be an ideal of A . Show that there exist ideals $I_1, I_2, \dots, I_n$ of R such that $I = I_1 \oplus I_2 \oplus \cdots \oplus I_n$ (*Note: This is not simply an isomorphism. It means that I_j is a subset of the j th summand of $R \oplus R \oplus \cdots \oplus R$.*)
 - Explicitly describe all prime ideals of A in terms of the prime ideals of R .
- Let R be a principal ideal domain and let K be the field of fractions of R . Let M and N be finitely generated R -modules. Let $\text{Hom}_R(M, N)$ be the set of R -module homomorphisms ϕ from M to N . It is an R -module: if $r \in R$, then $r\phi$ is defined by $(r\phi)(m) = r(\phi(m))$. Show that

$$\text{Hom}_R(M, N) \otimes_R K \simeq \text{Hom}_K(M \otimes_R K, N \otimes_R K)$$

as vector spaces over K .

CONTINUED ON NEXT PAGE

5. Let p and q be primes and let $K/\mathbb{Q}$ be a Galois extension of degree pq and Galois group G . Assume $p \equiv 1 \pmod{q}$.
- (a) Show that if G is non-abelian then there exists a polynomial $f(x) \in \mathbb{Q}[x]$ of degree p whose splitting field is K .
 - (b) Show that if G is abelian then there is no irreducible polynomial $f(x) \in \mathbb{Q}[x]$ of degree p whose splitting field is K .
6. Let G be a finite group and let H be a subgroup with $[G : H] = 2$. Let $\rho : G \rightarrow GL(V)$ be an irreducible representation of G (where V is some finite-dimensional vector space over $\mathbb{C}$). Suppose there exists a subspace $W \neq 0$ of V with $W \neq V$ and such that $\rho(h)W = W$ for all $h \in H$. Fix $g_0 \in G$ with $g_0 \notin H$. Let $W_0 = \rho(g_0)W$.
- (a) Show that $\rho(h)W_0 = W_0$ for all $h \in H$.
 - (b) Show that $W + W_0 = V$ and $W \cap W_0 = 0$.
 - (c) Let $\chi = \text{Trace}(\rho)$ be the character associated to ρ . Show that $\chi(g_0) = 0$.